

UDAY KIRAN KOTHAPALLI

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PROFESSIONAL EXPERIENCE

RFID Lab at Auburn University | Auburn, AL

Graduate Assistant

January 2024 – Present

- Contributed to a technical team integrating RFID technology with major retail self-checkout (SCO) systems, optimizing automated item identification and improving point-of-sale accuracy.
- Managed the end-to-end maintenance and troubleshooting of RFID hardware at live retail locations, ensuring high system uptime and adherence to strict safety and performance standards.
- Developed a data pipeline to ingest and process unstructured raw RFID telemetry from live retail environments, and trained deep learning models on this RFID time-series data to extract actionable insights for inventory movement and system efficiency.
- Communicated model results and technical findings to non-technical teammates and cross-functional stakeholders, translating complex machine learning outputs into clear, actionable insights.

Hyderabad City Police | Hyderabad, India

Cybersecurity Research Intern

October 2022 – January 2023

- Configured server infrastructures to support department-wide operations and data management and implemented the local network architecture across the department, ensuring high-speed connectivity and secure data transmission protocols.
- Analyzed raw financial datasets and bank transactions, leading to the identification and prevention of credit card fraud and identity theft cases.

Pradhi.ai | Hyderabad, India

Research Intern

September 2022 – February 2023

- Engaged in in-depth exploration of transformer models and subsequently applied BERT (Bidirectional Encoder Representations from Transformers) to analyze a legal corpus, demonstrating proficiency in leveraging advanced natural language processing techniques for legal text comprehension.

Spotverge | Hyderabad, India

Mobile Application Developer Intern

May 2022 – August 2022

- Contributed to the front-end development of an iOS/Android mobile application using Dart and Flutter.
- Implemented user data management with Google Firebase and ensured seamless data synchronization across platforms.

Cephas Research | Hyderabad, India

Remote Blockchain Developer Intern

May 2021 – August 2021

- Engineered a blockchain-based supply chain system designed to store product specifications and real-time GPS location data. Enhanced operational transparency and accountability within the supply chain process through immutable data recording.
- Developed technical proficiency in the Substrate framework and Rust programming language to build custom blockchain modules.

PUBLICATIONS

[P1] L. Medury, **U. K. Kothapalli**, and F. Kandah. "Towards Enhancing Device Anonymity and Classification Resistance in IoT," 2025 IEEE International Conference on Communications Workshops (ICC Workshops).

TECHNICAL SKILLS

Programming Languages: Python, Rust, C++, C, SQL, R, React, Dart

Machine Learning & AI: Deep Learning (LLM, Transformers), RAG, Computer Vision, Agentic AI.

Frameworks & Libraries: PyTorch, TensorFlow, Keras, Hugging Face Transformers, scikit-learn, NumPy, Pandas, OpenCV, LangChain, LlamaIndex.

Hardware: RFID Technology, Arduino Uno, Raspberry Pi

EDUCATION

Auburn University | Auburn, AL

Doctor of Philosophy in Computer Science and Software Engineering | Aug 2025 – Present

Auburn University | Auburn, AL

Master of Science in Cybersecurity Engineering | Aug 2023 – May 2025

TKR College of Engineering and Technology | Hyderabad, India

Bachelor of Technology in Computer Science and Engineering | Aug 2019 – June 2023